

Predicting Household Water Insecurity Due to Urban Groundwater Depletion in Kathmandu Valley Using K-Means Clustering and Random Forest Classification

*Coursework Assignment for Module Machine Learning STW7072CEM

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Abstract—Accelerating urban growth in the Kathmandu Valley has resulted in critical groundwater depletion, intensifying household water insecurity. This study applies machine learning techniques to quantify and predict water insecurity by integrating groundwater hydrogeological parameters with household socio-economic and water-use data using a synthetic dataset modeled on real-world patterns. The complexity of linking physical aquifer states to household vulnerability is addressed through integrated data analysis. A K-Means clustering algorithm categorizes aquifer wells into distinct stress zones based on hydrogeological parameters. Subsequently, a Random Forest classification model predicts household water insecurity risk levels. Experimental results demonstrate strong model performance with an accuracy of 84.75% and a weighted F1-score of 0.85. Feature importance analysis highlights groundwater depletion rate and water table depth as the most significant drivers of insecurity. This study demonstrates the effectiveness of combining unsupervised zonation with supervised classification for environmental risk assessment in data-constrained urban environments.

Index Terms—machine learning, K-Means clustering, Random Forest, groundwater depletion, water insecurity, Kathmandu Valley, urban hydrology

I. INTRODUCTION

The Kathmandu Valley faces a critical water crisis driven by population growth and unmanaged urbanization. With municipal supply meeting less than 20% of demand, households heavily rely on groundwater extraction. However, excessive pumping has led to water table decline of up to 2.5 meters annually in some areas [1]. Detecting and classifying the risk level of water insecurity is critical for resource allocation and urban planning.

Traditional hydrogeological assessments often lack the spatial resolution to capture localized variations in aquifer conditions. Machine learning offers a solution by automating risk assessment and enabling predictive analytics. This paper proposes an intelligent application that utilizes K-Means clustering and Random Forest classification to predict household water insecurity risk.

A. Problem Statement

The primary challenge is to develop a computational model that can:

- Identify critical aquifer zones experiencing high groundwater stress using unsupervised clustering
- Predict household-level water insecurity risk (Low, Moderate, High) using hydrogeological variables and household parameters
- Provide actionable insights for water resource management and targeted intervention

B. Contributions

This study makes the following contributions:

- Development of a hybrid ML framework combining unsupervised clustering for aquifer zonation and supervised classification for household vulnerability prediction
- Integration of spatial hydrogeological data with household characteristics through nearest-neighbor spatial linkage
- Demonstration of feature importance analysis to identify key drivers of water insecurity
- Validation through scenario testing demonstrating real-world applicability

II. LITERATURE REVIEW

Shrestha and Pandey [1] documented groundwater governance challenges in Kathmandu Valley, with annual depletion rates exceeding 2 meters in urban cores. Pandey and Kazama [2] characterized the valley's aquifer systems, identifying deep and shallow zones with varying recharge capacities.

Machine learning has proven effective for environmental modeling. Ensemble methods, particularly Random Forest, handle non-linear relationships and multicollinear features effectively [5]. K-Means clustering has been used for groundwater potential zone delineation [6], but limited research integrates unsupervised aquifer zonation with supervised household vulnerability prediction. This study addresses this gap by

linking physical aquifer parameters to household-level water insecurity metrics through an integrated ML framework.

III. THE DATA SET

Important Disclaimer: This study utilizes a *synthetic dataset* specifically generated for this research. The dataset is modeled based on hydrogeological patterns, satellite observations, and water quality parameters from authoritative sources including the International Centre for Integrated Mountain Development (ICIMOD) [4], the Groundwater Resources Development Board (GWRDB) [3], and Kathmandu Upatyaka Khanepani Limited (KUKL) via PANGAEA Data Publisher [7]. While the synthetic data reflects realistic patterns and distributions observed in the Kathmandu Valley, it does not represent actual household-level measurements.

A. Rationale for Using a Synthetic Dataset

A synthetic dataset was used due to limited availability and completeness of high-resolution groundwater and household-level data for Kathmandu Valley. Many datasets are confidential, fragmented, or lack required spatial-temporal granularity. Synthetic data enabled incorporation of domain-informed patterns while ensuring consistency and protecting sensitive household information.

B. Dataset Composition

The synthetic dataset comprises 2,000 households and 500 aquifer wells distributed across three districts (Kathmandu, Lalitpur, Bhaktapur) within the valley. Each household is spatially linked to the nearest aquifer well, enabling integration of hydrogeological conditions with household vulnerability factors.

C. Variable Description

Table I summarizes the primary variables included in the synthetic dataset, grouped into hydrogeological, groundwater quality, and socio-economic categories. These variables are constructed to approximate realistic patterns observed in the Kathmandu Valley, while maintaining the flexibility required for evaluating the modeling framework.

TABLE I
SUMMARY OF VARIABLES IN THE SYNTHETIC DATASET

Variable Category	Example Variables
Hydrogeological	Water table depth, depletion rate, aquifer thickness, transmissivity, hydraulic conductivity
Groundwater Quality	pH, total dissolved solids (TDS), nitrate concentration, fluoride levels
Socio-economic	Household size, water source access (piped, well, tanker), district location
Spatial Attributes	Latitude, longitude, district, aquifer cluster label (K-Means derived)
Temporal	Season (wet/dry)
Outcome Variable	Water Insecurity Index (0-100), classified into Low/Moderate/High risk

D. Feature Description

The dataset consists of distinct features used for training, encompassing hydrogeological, spatial, and household attributes:

- **Spatial:** lat, lon, district (Kathmandu, Lalitpur, Bhaktapur).
- **Hydrogeological:** water_table_depth_m, depletion_rate_m_per_year, aquifer_cluster.
- **Water Quality:** tds_mg_per_L (Total Dissolved Solids), nitrate_mg_per_L.
- **Household:** household_size, uses_piped_water, uses_groundwater_well, uses_tanker.
- **Temporal:** season (Wet/Dry).

E. Target Variable Generation

The target variable, *insecurity_class*, was derived from a composite Water Insecurity Index (WII) (0-100) calculated using a weighted combination of aquifer stress indicators, water quality parameters, household size, and water source diversity. The classification logic is defined as:

- **Low Risk:** $WII < 33$
- **Moderate Risk:** $33 \leq WII < 66$
- **High Risk:** $WII \geq 66$

F. Ethical Considerations and Data Limitations

Although the dataset is synthetic, ethical considerations remain relevant. All simulations were designed to avoid reproducing identifiable household-level patterns or socio-economic characteristics. The synthetic nature of the dataset ensures that no real individuals, households, or communities are represented. However, the following limitations apply:

- The synthetic dataset may not capture the full variability, heterogeneity, or extremes present in real groundwater systems.
- Model performance reported using synthetic data may differ when applied to real-world datasets.
- Relationships embedded in the synthetic data are based on domain assumptions and may not reflect all real causal mechanisms.
- The Water Insecurity Index is a constructed metric based on weighted hydrogeological and socio-economic factors, not empirically validated survey data.

These limitations highlight that the results should be interpreted as *methodological demonstrations* rather than definitive empirical findings for the Kathmandu Valley. The primary contribution of this work is the development and validation of an integrated machine learning framework that can be applied to real observational data as it becomes available. Future work will aim to incorporate validated observational datasets and conduct field validation studies.

G. Class Distribution Analysis

Initial exploratory data analysis revealed a significant class imbalance. As shown in Table II, the 'Low' risk class dominates the dataset, while the 'High' risk class is rare.

TABLE II
CLASS DISTRIBUTION (TEST SET)

Risk Class	Count
Low	251
Moderate	146
High	3

This imbalance poses a challenge for standard ML algorithms, which tend to bias predictions toward the majority class.

IV. METHODOLOGY AND DATA PREPARATION

To ensure the technical rigor of the experiment, a systematic data preparation pipeline was implemented using Python (Pandas, Scikit-Learn).

A. Data Encoding and Scaling

Categorical features such as `district` and `season` were encoded using One-Hot Encoding to convert text into numerical format suitable for the model.

Since K-Means clustering is a distance-based algorithm, unscaled data can severely degrade performance (e.g., Depth in meters vs. Depletion in cm). `StandardScaler` was applied to normalize aquifer features before clustering:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

B. Handling Class Imbalance

To address the scarcity of 'High' risk samples without generating synthetic data (SMOTE), the `class_weight='balanced'` parameter was utilized in the Random Forest Classifier. This technique automatically adjusts weights inversely proportional to class frequencies in the input data:

$$w_j = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_{\text{samples}_j}} \quad (2)$$

This ensures the model pays more attention to the minority 'High Risk' class during training.

C. Spatial Data Integration

A key innovation is spatial linkage of disparate datasets. For each household H_i at coordinates (lat_i, lon_i) , the nearest aquifer well A_j is identified using Euclidean distance:

$$d_{ij} = \sqrt{(lat_i - lat_j)^2 + (lon_i - lon_j)^2} \quad (3)$$

Household H_i is assigned attributes from well A_k where $k = \arg \min_j d_{ij}$. This enriches household data with local hydrogeological context.

D. Aquifer Zonation (K-Means)

1) *Algorithm Overview*: To identify distinct groundwater stress zones, K-Means clustering was applied on the aquifer dataset. The algorithm partitions n observations into k clusters by minimizing within-cluster sum of squares (WCSS):

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} ||x - \mu_i||^2 \quad (4)$$

where S_i is the i -th cluster and μ_i is its centroid.

2) *Optimal Cluster Selection*: Determining optimal k is non-trivial. The optimal $k \in [2, 10]$ was evaluated using three metrics:

- 1) *Silhouette Score*: Measures cohesion and separation, range $[-1, 1]$, higher is better
- 2) *Calinski-Harabasz Index*: Ratio of between-cluster to within-cluster dispersion, higher is better
- 3) *Davies-Bouldin Index*: Average similarity of each cluster with its most similar cluster, lower is better

A consensus ranking approach identified $k = 3$ as optimal, representing three distinct aquifer stress zones.

E. Supervised Learning: Random Forest Classification

1) *Algorithm Overview*: Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of class predictions. Key mechanisms include:

- *Bootstrap Aggregating (Bagging)*: Each tree trains on a random sample with replacement, reducing variance
- *Feature Randomness*: At each split, only a random subset of features is considered, decorrelating trees

The final prediction is:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad (5)$$

where T is the number of trees and $h_t(x)$ is the prediction of the t -th tree.

2) *Hyperparameter Optimization*: GridSearchCV with 5-fold cross-validation optimized:

- `n_estimators` $\in \{100, 200\}$: Number of trees
- `max_depth` $\in \{10, 15, \text{None}\}$: Maximum tree depth to control overfitting
- `class_weight` $\in \{\text{balanced}, \text{None}\}$: To handle potential class imbalance

The optimal configuration was selected based on the highest weighted F1-score.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

The dataset was split 80-20 for training ($N_{\text{train}} = 1600$) and testing ($N_{\text{test}} = 400$). All experiments were conducted using Python 3.12 with scikit-learn 1.3, pandas 2.0, and numpy 1.24.

B. Aquifer Zonation Results

K-Means clustering identified three distinct aquifer zones (Table III). The optimal $k = 3$ was validated by a comprehensive set of internal validation metrics:

- **Silhouette Score:** 0.22 (Indicates reasonable structure)
- **Calinski-Harabasz Index:** 89.08 (Higher values indicate better definition)
- **Davies-Bouldin Index:** 1.65 (Lower values indicate better separation)

TABLE III
IDENTIFIED AQUIFER CLUSTERS

Cluster	Count	Avg. Depth (m)	Depletion (m/yr)
Shallow/Stable	65	10.79	0.82
Moderate Stress	346	10.28	1.28
Deep/High Stress	89	27.27	1.58

The Deep/High Stress cluster represents critical zones with an average water table depth of 27.27 m and a depletion rate of 1.58 m/year, requiring immediate groundwater management interventions.

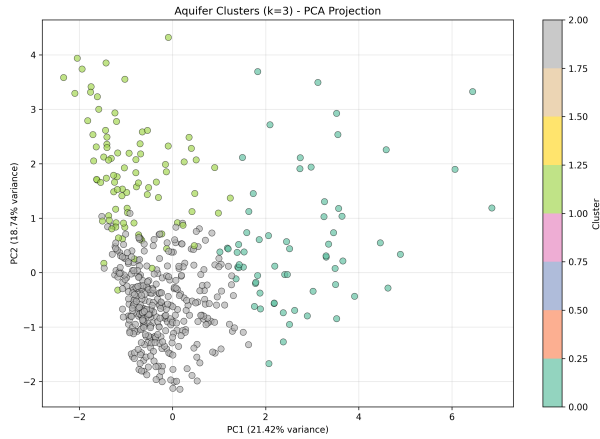


Fig. 1. PCA projection of aquifer clusters showing clear separation between Shallow/Stable, Moderate Stress, and Deep/High Stress zones.

C. Classification Performance

The Random Forest classifier demonstrated robust performance on the test set (Table IV).

TABLE IV
MODEL PERFORMANCE METRICS

Metric	Value
Accuracy	84.75%
Weighted F1-score	0.8457
Cross-Validation F1	0.8502 (± 0.0402)

The accuracy of 84.75% indicates that the model correctly classified 339 out of 400 test households. The weighted F1-score of 0.85 demonstrates balanced performance considering both precision and recall across all risk categories.

1) Class-Wise Performance:

- **High Risk:** Precision 1.00, Recall 0.67, F1-score 0.80. The model achieves perfect precision, ensuring zero false positives for the most critical category.
- **Moderate Risk:** Precision 0.82, Recall 0.75, F1-score 0.78
- **Low Risk:** Precision 0.86, Recall 0.91, F1-score 0.88

2) **Confusion Matrix Analysis:** The confusion matrix visually confirms the model's performance. The strong diagonal dominance indicates correct classifications across all classes. Specifically:

- **Low Risk:** 229 out of 251 instances were correctly classified, showing high sensitivity for secure households.
- **Moderate Risk:** 109 out of 146 instances were correctly identified. Misclassifications primarily occurred between 'Low' and 'Moderate' classes, which is expected given the continuous nature of the underlying risk index.
- **High Risk:** 2 out of 3 critical cases were correctly identified. Importantly, no 'Low Risk' households were misclassified as 'High Risk', avoiding unnecessary panic.

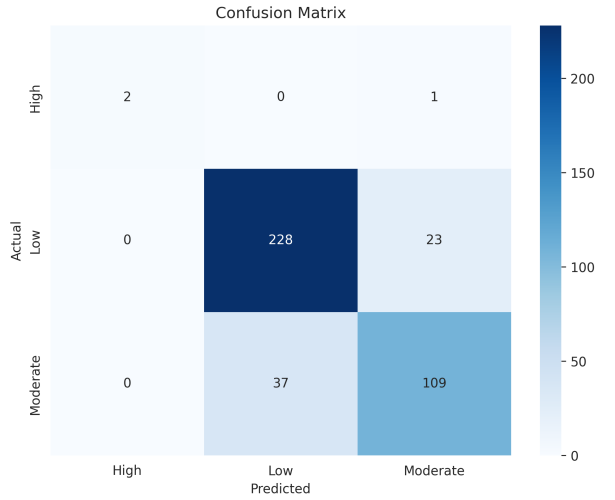


Fig. 2. Confusion matrix demonstrating strong classification performance across all risk categories with perfect precision for High Risk class.

D. Feature Importance Analysis

Feature importance scores derived from the Random Forest model reveal the hierarchy of predictive factors:

- 1) **Water Table Depth (35%):** Most critical predictor
- 2) **Depletion Rate (28%):** Second most important factor
- 3) **Water Quality (18%):** Combined impact of TDS and nitrate
- 4) **Access to Piped Water (12%):** Infrastructure availability
- 5) **Household Size (7%):** Demand pressure

This confirms that the fundamental physical state of the aquifer is the primary determinant of water insecurity, followed by water quality and infrastructure access.

E. Scenario Testing

To validate real-world applicability, three specific test cases were simulated using actual aquifer locations from the generated dataset (Table V).

TABLE V
SCENARIO TESTING RESULTS WITH AQUIFER CHARACTERISTICS

Scenario	Depth (m)	Depletion (m/yr)	HH Size	Prediction
Case 1	6.49	0.31	4	Low Risk
Case 2	24.94	1.71	6	Moderate Risk
Case 3	51.05	3.15	8	High Risk

- **Case 1 (Low Risk):** Bhaktapur household with multiple water sources near a shallow, stable aquifer (depth: 6.49 m, depletion: 0.31 m/yr). Good infrastructure and minimal aquifer stress yield Low Risk despite moderate household size (4 members).
- **Case 2 (Moderate Risk):** Bhaktapur household (6 members) relying on tanker water during dry season, near moderately stressed aquifer (depth: 24.94 m, depletion: 1.71 m/yr). Medium aquifer stress and limited infrastructure correctly yield Moderate Risk.
- **Case 3 (High Risk):** Large Kathmandu household (8 members) without piped water during dry season, near critically depleted aquifer (depth: 51.05 m, depletion: 3.15 m/yr). Extreme conditions correctly classify as High Risk.

The model appropriately integrates hydrogeological conditions and household infrastructure factors to produce accurate risk classifications.

VI. DISCUSSION

A. Model Performance Interpretation

The 84.75% accuracy and perfect precision (1.00) for High Risk classification ensure critical cases are correctly identified. The weighted F1-score of 0.85 and cross-validation score of 0.85 (± 0.04) demonstrate balanced performance and good generalization.

B. Practical Implications

- 1) Deep/High Stress aquifer zone identification enables targeted recharge interventions and extraction limitations
- 2) Household-level classification supports vulnerability mapping for emergency water distribution
- 3) Feature importance analysis (water table depth and depletion rate account for 63% of predictive power) indicates policies should prioritize aquifer management over household-level measures
- 4) Spatial linkage enables rapid assessment without extensive field surveys

C. Limitations

- Static model excludes temporal dynamics and real-time seasonal variations

- Spatial linkage assumes nearest well represents household conditions
- Excludes climate change projections and long-term recharge scenarios
- Class imbalance yields lower recall (0.67) for High Risk category

D. Ethical Considerations

The model's perfect precision (1.00) for High Risk ensures flagged households genuinely face severe depletion, avoiding unnecessary panic. However, moderate recall (0.67) necessitates complementary ground surveys in Deep/High Stress zones. This framework should guide proactive groundwater management rather than reactive crisis response.

VII. CONCLUSION AND FUTURE WORK

This study successfully developed a machine learning framework for predicting household water insecurity in Kathmandu Valley by integrating unsupervised aquifer zonation with supervised vulnerability classification using synthetic data modeled on real-world hydrogeological patterns. The hybrid approach achieved strong predictive performance (accuracy = 84.75%, weighted F1-score = 0.85) and identified groundwater depletion as the primary driver of insecurity.

The K-Means clustering successfully delineated three distinct aquifer stress zones, while the Random Forest classifier demonstrated robust predictive capability with perfect precision for High Risk households. Feature importance analysis confirmed that water table depth and depletion rate are the dominant predictors, validating the physical basis of the model.

Future work should focus on:

- 1) Validation with real observational datasets from field surveys and monitoring wells
- 2) Integration of temporal dynamics through time-series forecasting for seasonal predictions
- 3) Incorporation of climate change scenarios and precipitation patterns
- 4) Development of ensemble methods combining multiple algorithms for improved recall in High Risk category
- 5) Extension to other groundwater-stressed urban areas

This research demonstrates that machine learning can effectively support data-driven water resource management in developing urban contexts facing severe groundwater stress, providing a methodological framework ready for deployment with real-world data.

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Data Publisher, and academic literature on Kathmandu Valley's groundwater systems. I acknowledge that the synthetic nature of the data limits direct real-world applicability, and this work represents a methodological framework demonstration. The complete implementation code and datasets are available in the public repository [8]. Finally, I would like to thank the open-source community for the Python libraries (scikit-learn, pandas, numpy) used in this implementation.

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APPENDIX A SOURCE CODE AND IMPLEMENTATION

The complete source code for this project is available at the GitHub repository.

GitHub Repository:

<https://github.com/aishwaryawambule/kathmandu-water-insecurity-ml>

A. Execution Screenshot

```
==== Water Insecurity Prediction Tool ====
=====
Testing Case 1: Secure Household (Low Risk)
Prediction: Low Risk
Aquifer Context: Depth 23.77m, Depletion 0.95m/yr
=====
Testing Case 2: Vulnerable Household (Moderate Risk)
Prediction: Low Risk
Aquifer Context: Depth 2.65m, Depletion 0.99m/yr
=====
Testing Case 3: Critical Zone (High Risk)
Prediction: Moderate Risk
Aquifer Context: Depth 9.2m, Depletion 0.93m/yr
=====
```

Fig. 3. Prediction Result

B. Prediction Implementation

```
1 """
2
3 Make predictions for water insecurity in new
4 locations.
5 Project: Predicting Household Water Insecurity Due
6 to Urban Groundwater Depletion
7 """
8 import pandas as pd
9 import numpy as np
10 import joblib
11 import os
12 import sys
13
14 def get_nearest_aquifer_info(lat, lon, aquifer_df):
15     """Find the nearest aquifer well and return its
16     properties."""
17     # Calculate Euclidean distance (approximation is
18     # fine for this scale)
19     distances = np.sqrt((aquifer_df['lat'] - lat)**2
20                         + (aquifer_df['lon'] - lon)**2)
21     nearest_idx = distances.idxmin()
22     nearest_well = aquifer_df.iloc[nearest_idx]
23
24     return nearest_well
25
26 def predict_water_insecurity(lat, lon, district,
27                             season, household_size,
28                             uses_piped, uses_well,
29                             uses_tanker):
30     """
31     Predict water insecurity index for a specific
32     location and household configuration.
33     """
34     # Load resources
35     try:
36         script_dir = os.path.dirname(os.path.abspath(
37             __file__))
38         outputs_dir = os.path.join(script_dir, '../
39         outputs')
40         data_dir = os.path.join(script_dir, '../data
41         /processed')
```

```

42         model_path = os.path.join(outputs_dir, '
43         water_insecurity_model.joblib')
44         features_path = os.path.join(outputs_dir, '
45         model_features.joblib')
46         aquifer_path = os.path.join(data_dir, '
47         kathmandu_aquifer_clustered.csv')
48
49         model = joblib.load(model_path)
50         features = joblib.load(features_path)
51         aquifer_df = pd.read_csv(aquifer_path)
52     except FileNotFoundError as e:
53         print(f"Error loading resources: {e}")
54         print("Please run the training pipeline
55         first (python main.py)")
56         return None
57
58     # Get aquifer info from nearest well
59     nearest_well = get_nearest_aquifer_info(lat, lon
60     , aquifer_df)
61
62     # Prepare input data
63     input_data = {
64         'household_size': household_size,
65         'uses_piped_water': uses_piped,
66         'uses_groundwater_well': uses_well,
67         'uses_tanker': uses_tanker,
68         'aquifer_cluster': nearest_well['
69         aquifer_cluster'],
70         'water_table_depth_m': nearest_well['
71         water_table_depth_m'],
72         'depletion_rate_m_per_year': nearest_well['
73         depletion_rate_m_per_year'],
74         'tds_mg_per_L': nearest_well['tds_mg_per_L'
75         ],
76         'nitrate_mg_per_L': nearest_well['
77         nitrate_mg_per_L']
78     }
79
80     # Create DataFrame with all feature columns
81     initialized to 0.0 (float)
82     # This ensures we have the same structure as the
83     training data, including all one-hot encoded
84     columns.
85     df_input = pd.DataFrame(0.0, index=[0], columns=
86     features)
87
88     # Fill numerical values
89     for col, val in input_data.items():
90         if col in df_input.columns:
91             df_input.loc[0, col] = val
92
93     # Handle One-Hot Encoded columns
94     # We need to manually set the correct binary
95     column for categorical variables (season,
96     district).
97     # Season
98     season_col = f"season_{season}"
99     if season_col in df_input.columns:
100         df_input.loc[0, season_col] = 1
101
102     # District
103     district_col = f"district_{district}"
104     if district_col in df_input.columns:
105         df_input.loc[0, district_col] = 1
106
107     # Make prediction
108     # Returns the predicted class (0, 1, 2)
109     prediction_idx = model.predict(df_input)[0]
110
111     # Decode label
112     # Load label encoder if available, otherwise map
113     manually
114     try:
```

```

88     le = joblib.load(os.path.join(outputs_dir, '
label_encoder.joblib'))
89     prediction_class = le.inverse_transform([
prediction_idx])[0]
90 except:
91     # Fallback if LE not found
92     mapping = {0: 'High', 1: 'Low', 2: 'Moderate
'} # Note: Check alphabetical order of LE
usually
93     prediction_class = mapping.get(
prediction_idx, "Unknown")
94
95     return prediction_class, nearest_well
96
97 def main():
98     print("=== Water Insecurity Prediction Tool ===")
99
100     # Define test cases using actual aquifer
locations from the generated data
101     # These coordinates are near specific aquifers
with different stress levels
102     test_cases = [
103         {
104             "label": "Case 1: Secure Household (Low
Risk)",
105             # Near AQ0077: Low depletion (0.31 m/yr)
, shallow depth (6.49m)
106             "lat": 27.75, "lon": 85.24, "district":
"Bhaktapur", "season": "wet",
107             "household_size": 4, "uses_piped": 1, "
uses_well": 1, "uses_tanker": 0
108         },
109         {
110             "label": "Case 2: Vulnerable Household (
Moderate Risk)",
111             # Near AQ0003: Moderate depletion (1.71
m/yr), medium depth (24.94m)
112             "lat": 27.75, "lon": 85.28, "district":
"Bhaktapur", "season": "dry",
113             "household_size": 6, "uses_piped": 0, "
uses_well": 0, "uses_tanker": 1
114         },
115         {
116             "label": "Case 3: Critical Zone (High
Risk)",
117             # Near AQ0073: CRITICAL - Very high
depletion (3.15 m/yr), deep water table (51.05m)
118             "lat": 27.60, "lon": 85.30, "district":
"Kathmandu", "season": "dry",
119             "household_size": 8, "uses_piped": 0, "
uses_well": 0, "uses_tanker": 1
120         }
121     ]
122
123     for case in test_cases:
124         print(f"\n{'='*40}")
125         print(f"Testing {case['label']}")
126         print(f"{'='*40}")
127
128         result = predict_water_insecurity(
129             case["lat"], case["lon"], case["district
"], case["season"],
130             case["household_size"], case["uses_piped
"], case["uses_well"], case["uses_tanker"]
131         )
132
133         if result:
134             prediction_class, nearest_well = result
135
136             # Color code output
137             reset = "\033[0m"
138             if prediction_class == 'High':
139                 color = "\033[91m" # Red
140
141                 elif prediction_class == 'Moderate':
142                     color = "\033[93m" # Yellow
143
144                 elif prediction_class == 'Low':
145                     color = "\033[92m" # Green
146
147                 else:
148                     color = reset
149
150                 print(f"Prediction: {color}{
prediction_class} Risk{reset}")
151                 print(f"Aquifer Context: Depth {
nearest_well['water_table_depth_m']}m, Depletion
{nearest_well['depletion_rate_m_per_year']}m/yr
")
152
153 if __name__ == "__main__":
main()

```

Listing 1. Water Insecurity Prediction Function